Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2024	ENGI3451-WE01

Title: (BEng/MEng)

Electronics and Communications 3

Time Allowed:	2 hours
Additional Material provided:	None
Materials Permitted:	N/A
Calculators Permitted:	Those from the Casio fx-83 and fx-85 series.

Instructions to Candidates:	This is an in-person, invigilated exam. Answer ALL questions. Write your answers in the white-covered answer booklet with barcodes. Begin your answer to each question on a new page. All relevant workings must be shown.

Revision:	

Section A

Question 1

(a) Consider a carrier signal $c(t) = \frac{7}{4}\cos(9 \times 10^5 \pi t)$ and the periodic message signal shown in Figure Q1. Assume double-sideband (DSB) amplitude modulation.

(i) Determine the modulation index.

(ii) Determine the power efficiency.

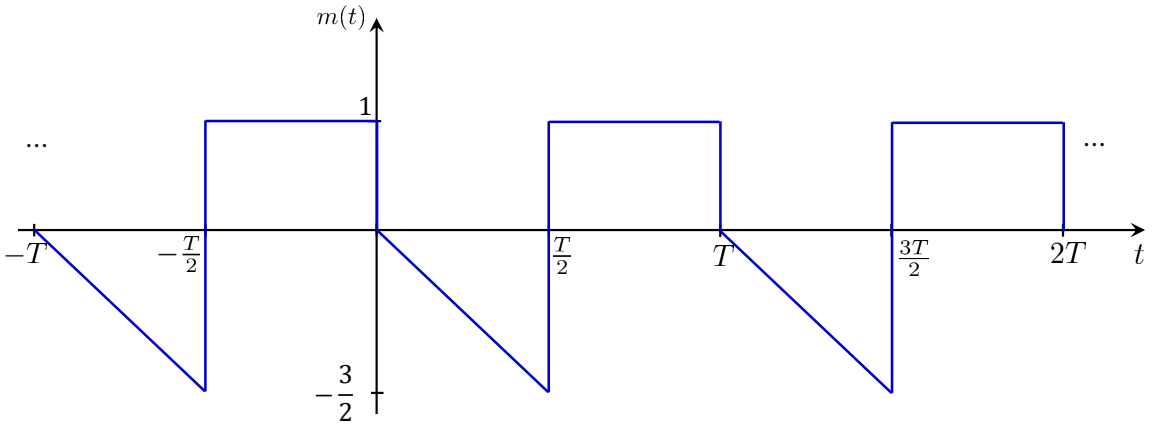


Figure Q1: A message signal.

[40%]

(b) Assume double-sideband (DSB) amplitude modulation (AM). The envelope detector is widely used for demodulation of DSB-AM signals. The input signal to the envelope detector is: $s(t) = [3 + 2\cos(3\pi 10^3 t)]\cos(8\pi 10^5)$.

(i) Draw the envelop detector circuit.

(ii) Explain how the envelope detector circuit works.

(iii) Determine the optimum value of the time constant RC to ensure a good tracking of the envelope.

[50%]

(c) Describe three main disadvantages of double-sideband (DSB) amplitude modulation (AM).

[10%]

continued

Question 2

(a) A discrete memoryless source produces outputs s_1, s_2, s_3, s_4 and s_5 with probabilities 0.4, 0.2, 0.15, 0.1 and 0.15, respectively. Assume coding over an alphabet of size $D = 4$ (quaternary). Note that a quaternary code is one which uses four alphabets in coding, e.g., $\{0, 1, 2, 3\}$.

(i) Design a Huffman code for this source.

(ii) Determine the average codeword length.

(iii) Determine the entropy of this source. Comment on the result.

[45%]

(b) Consider a wireless communication system with a band-limited additive white Gaussian noise (AWGN) channel with bandwidth $W = 20$ MHz and a signal-to-noise ratio (SNR) of 0 dB.

(i) Determine the capacity of this channel in bits/s.

(ii) Determine the minimum SNR, in dB, required so that the transmitter can transmit with a rate of $R = 60$ Mbps reliably.

[30%]

(c) The generator matrix for a linear binary code is

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

(i) What is the rate of this code?

(ii) Determine if 111011 is a valid codeword for this code.

[25%]

continued

Section B

Please note that in this section mathematical answers will only be marked as correct if the correct units are also included with the answer.

Question 3

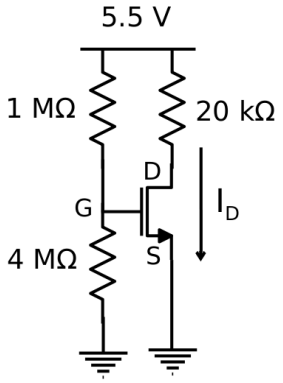
(a) A MOSFET has a turn on voltage of 0.5 V, a channel width of 10 μm and length of 2 μm. The mobility of the material system is 100 cm^2/Vs and the capacitance of the oxide layer is 1nF. Calculate beta of the transistor and the current it will produce if a gate voltage of 2.0 V is applied if the device is in the saturation region.

[20%]

(b) Draw typical drain-source voltage v.s. drain-source current curves for a MOSFET with a turn on voltage of 1.0 V operating with V_{GS} at around 4 V. Make sure you include the effect of channel pinching by including V_m on your graph. Write down the equation relating V_m to the drain current assuming the device is in saturation.

[20%]

(c) You are reverse engineering a radio transmitter. You find it has the following circuit in it:



Calculate ID I_D assuming the transistor has a β of 2 mA/V^2 and a turn on voltage of 1 V. Show all the steps of your working.

[30%]

(d) You are designing an amplifier using a MOSFET and want to find out if the gate source capacitance will limit your circuit's frequency performance. Derive the equation relating transconductance and CGS to calculate the maximum frequency response for a MOSFET. Show using an equation how the frequency response of your device could be increased without changing its dimensions.

[30%]

continued

Question 4

(a) You are designing an emergency warning system for a nuclear power station. The circuit should monitor the temperature in a cooling pool where radioactive waste is stored. As part of this system you need to design a differential amplifier. Draw a differential amplifier circuit attached to a load and label the transistor providing the current mirror. Give two reasons why current mirrors are useful.

[20%]

(b) A temperature sensor in the cooling pool of the reactor produces a two bit binary number (encoded a, b) which signifies the temperature being measured. If the temperature in the cooling pool goes over 75 degrees a siren should sound, the siren sounds when its input Q is low (zero). A truth table representing the problem is shown below:

Temperature	a	b	Q
0-25	0	0	1
25-50	0	1	1
50-75	1	0	1
$>$75	1	1	0

(i) Write the minimal boolean expression for this truth table.

(ii) Design and draw out a digital circuit using MOSFETs which comprises of a pull-up and pull-down circuit to implement the truth table.

[30%]

(c) (i) As part of your monitoring system you need to design an oscillator circuit. Draw out the circuit diagram of a phase shift oscillator and select components to give an oscillation of 1 kHz. As part of the design specification you only have 10 kOhm resistors available, but you are free to choose any value of capacitance you want.

(ii) What could be a potential problem with the spectrum produced by this circuit? Name two ways that this could be overcome. Sketch the gain v.s. voltage response of an improved circuit.

[50%]

END